

MSM538032E@SOP44.BIN analysis report

Executive identification

- Chip/dump size: OKI / LAPIS MSM538032E, 8-Mbit mask ROM, 1 MiB as x8; file size 1,048,576 bytes / 0x100000.
- Likely role: IBM PC110 Japanese/System font mask ROM; first 8 KiB is a valid PC option-ROM font window.
- Confidence: Very high.
- SHA-256: 9829fdb8281c12022dc3b77686044ed1a5213ab526ce4329f2841cd64171784c
- MD5: 8a160b1ad6dc7a64c086456eaa4e54a6
- Byte entropy: 5.9395 bits/byte.

Evidence from this dump

- Starts with 55 AA 10 CB; byte 2 is 0x10, so the option-ROM window length is 16 * 512 = 8192 bytes.
- Checksum over the first 8192 bytes is 0 modulo 256, so the first-window option-ROM header is valid.
- Header/catalog strings include FONT, 84G7940, IBM copyright, 03/23/95, and System SBCS/DBCS font size descriptors.
- Most of the remaining 1 MiB image is bitmap-like data with repeated glyph patterns, not executable code.

Structural notes

- Active 4 KiB ranges (heuristic, excludes mostly 0x00/0xFF blocks): 0x00000-0x09FFF, 0x0B000-0x55FFF, 0x59000-0xCAFFF, 0xCC000-0xD2FFF, 0xD6000-0xFFFFF
- Longest 0xFF runs: 0x3080 len 256, 0x47080 len 256, 0xD8C80 len 256, 0x31AC len 60, 0x471AC len 60
- Longest 0x00 runs: 0x562C1 len 13764, 0xD2FF0 len 12744, 0xCA97F len 6344, 0x9E1C len 6118, 0x415C0 len 5664
- PC option-ROM signatures found:
 - 0x000000: size byte 0x10 (8192 bytes), checksum mod 256 0, valid=True, header 55 aa 10 cb 00 00 00 00
 - 0x03C8F8: size byte 0x54 (43008 bytes), checksum mod 256 157, valid=False, header 55 aa 54 aa 51 ff 40 00
 - 0x03C918: size byte 0x54 (43008 bytes), checksum mod 256 230, valid=False, header 55 aa 54 aa 51 ff 40 00

Interesting strings

- 0x00000E: FONT
- 0x000080: 84G7940 (C) Copyright IBM Corporation 1990, 1995 All Rights Reserved
- 0x000210: System SBCS 16
- 0x000240: System SBCS 19
- 0x000270: System DBCS 16 -
- 0x0002A0: System SBCS 24
- 0x0002D0: System DBCS 24 -
- 0x000300: System SBCS 12
- 0x000330: System DBCS 12 -
- 0x03CD58: =VeR

Included supplemental files

- strings/MSM538032E_font_maskrom.strings.txt - all printable strings found by a simple ASCII scanner.

- `hexdumps/MSM538032E_font_maskrom_selected_hexdumps.txt` - selected annotated hexdumps.
- `disassembly/MSM538032E_font_maskrom_font_window_catalog.txt`

Caveats

- This is static triage, not a full reverse-engineering project. Hardware register names, MCU peripheral vector names, and exact memory maps require board schematics, data books, or emulator instrumentation.